# Electric Vehicle Adoption Trends and Consumer Sentiment Analysis

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***Abstract*—This study investigates the adoption of electric vehicles (EVs) using a dual-faceted approach: prediction of sales and consumer sentiment analysis. In more detail, the quantitative methods are forecast models like the Bass Diffusion Model and its derivatives (Generalized and Uncertain Bass Models) that estimate the tendency of adoption relying on historical records of sales. Consumer sentiment is the use of machine learning and deep learning models to assess the emotions of consumers. This has been evidenced in ev adoption results that demonstrate the continuous increase in growth patterns occasioned by government incentives, market forces, and shifting consumer choice. Charging infrastructure and product affordability are thus vital issues to address in order to sustain the level of growth.**

***Index Terms*—Electric Vehicle Adoption, EV Sales Forecasting, Bass Diffusion Model, Sentiment Analysis, Machine Learning in EV Market, Consumer Preferences, EV Sentiment Distribution, Deep Learning for Sentiment Classification, Time-Series Analysis, EV Market Trends, Sustainable Transportation.**

## I. Introduction

The transport sector is going through a transition to electric vehicles (EVs) on account of environmental improvement and curbing greenhouse gas emissions. The use of EV"s has been embraced globally due to the lowered Co2 emissions and improved energy efficiency that it takes to the fight against climate change. Governments globally have understood this potential, they have applied incentives like subsidies and tax credits for the same. For instance, over-reliance of the electrical vehicle or firm policy support has seen many nations achieve rapid sales growth over the last decade.

Despite , there are still challenges in the way of introducing the technology to mainstream use, such as, firstly, high initial costs; secondly, insufficient number of recharging stations; and, thirdly, consumer acceptance about its stability in the long-term usage. Such challenges are even more apparent in emerging markets where other accompanying structures and frameworks are comparatively less developed. These concerns, and the drivers for and against adoption, are critical to the long-term sustainment of growth in the EV market.

To overcome these challenges this study will use two converging research approaches. First, it uses sophisticated accuracy forecasts models, such as the Bass Diffusion Model, which is used to explain the patterns of innovation and imitation, hence, used to adopt new technologies. Second, it uses the machine learning sentiment analysis to analyze and interpret the perception and concern that consumers have for EV"s. Employing both analytical methods, this research seeks to offer recommendations to policymakers and manufacturers that will improve their understanding of the changing nature of the EV market.

And The change towards electric vehicle (EV) is one of the most significant revolution witnessed in the transport industry due to the rising demand of sustainable energy and technology. They are the answer to the increasing pollution and rigorous control over emissions of green house gases EVs are the future in the fight against climate change. In the last ten years, improvement in battery technology, decreased costs of production and favorable government policies to the use of electric vehicles have enhanced the uptake of EVs. Measurable programs like subsidies, tax exemption, and infrastructure development exemplify the readiness of the government and industries towards transforming to sustainability.

There are many fundamental parameters that have been driving the market for electric vehicles, including technological progress and policy actions that continue to act as a solid supporting wall for their increased implementation. There are social factors that relate to EVs that include, price, efficiency, recharging technology, and the ecological effect that shape the buying behavior. For instance, although customers recognize the advantages of using EVs in minimizing environmental impacts of the society some of the issues realise people to accept them for instance, range of the car when fully charged, time taken to charge among other issues on the battery life. This paper established such sentiments for automakers and policymakers to identify such hurdles and improve customer confidence.

However, the road map towards total EV penetration is full of challenges as this section will show. This results in charging infrastructure lacking in many parts of the country, particularly rural areas leading to inequalities between urban and rural areas. However, the cost of EV is still relatively high as compared to ICE vehicles, and is keep on discouraging many buyers. As regards the environmental context, the creation and the disposal of EV batteries raise new challenges for sustainability that have to be met. Such concerns highlight the sparse theoretical studies needed to establish more profound problems that determine the promotion of EVs.

This research aims to bridge the gap between technological

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advancements and consumer behavior by integrating data-driven insights and sentiment analysis. Through the application of diffusion models to determine future sales of EVs and the use of machine learning in analyzing the customer feedback this study offers the grand theory of the EV market. Besides counting the numbers of adopting consumers, this is too able to identify issues relating to consumer satisfaction and concern that may be prompting the adoption. The result may useful for the decision maker such as automaker, the policy maker and the environmentalist for formulating the right policies and strategies to drive EV adoption that optimally improves the customer experience.

Naturally, in such setting which appear to be experiencing frequent changes constantly, it is very important to differentiate more between the market approaches and customer attitudes. This paper aims at shedding light on the feasibility and implications for the transition to electric cars in order to pave way for the creation of practical solutions aimed at enhancing consumption and sustainability in this sector.

## II. Review of Literature

This research integrates insights from various studies focusing on:

### *A. Diffusion Models: Bass Diffusion and Generalized Bass Models Predicting EV Adoption*

Global Ratios of battery electric vehicles When undertaking comparisons of the ratios of different countries in adoption of battery electric vehicles, this research has noted that all the conditions present exponential growth. If one bases his/her estimates on modeling, then there is a propensity for system-wide adoption that appears significantly faster than the potential extraordinary analysis of economics have predicted. [3]

Prediction of PEV Adoption with Agent-Based Parameterized Bass Network Diffusion Model: This research develops an agent-based parameterized Bass network diffusion model for EV population data, achieving high estimation accuracy for EV adoption in both temporal and geographical scopes. [2]

### *B. Sentiment Analysis: Machine Learning Models and Deep Learning Techniques (e.g., LSTM) Analyzing Textual Data*

Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery, 2018.This survey provides an overview of deep learning methods in sentiment analysis, discussing various models and their effectiveness in capturing sentiment from text data. [4]

But then, as we have seen, the baseline approach using traditional powered machine learning models is still useful for sentiment classification while the LSTMs and Transformers add a new dimension to the domain of deep learning. [4] [5]

Two important studies provide deeper insights into electric vehicle (EV) adoption and its implications:

### *C. "Prediction of Electric Vehicle Penetration and Its Impacts on Distribution Systems: A Real-World Case Study in Maryland"*

**Source:** UCR ECE INTRANET

This study introduces an integrated data-driven planning framework designed to assist electric utilities in forecasting EV adoption and assessing its effects on distribution feeders. The framework comprises two primary modules:

- **Generalized Bass Diffusion Model (GBM):** Utilizes historical adoption data, EV availability, incentives, costs, and demographic information to predict EV adoption at granular levels, such as zip codes or specific feeders.
- **Impact Analysis Module:** Combines EV adoption forecasts with representative EV charging load profiles to evaluate potential impacts on distribution feeders, including voltage violations and equipment overloading.

The framework was validated through a case study in Maryland, employing real-world data and distribution circuit models. Findings indicate that the GBM effectively predicts EV adoption patterns, enabling utilities to identify when and where infrastructure upgrades are necessary to accommodate increasing EV penetration.

### *D. "Development of a Generalization Bass Diffusion Model for Chinese Electric Vehicle Market with the Consideration of Charging Infrastructure"*

**Source:** IEEE XPLORE

This study is based on a case of China's EV market to arrive at a Generalized Bass Diffusion Model that incorporates the availability of charging facilities as a critical factor in the uptake of EVs. Selling patterns were used to establish the relationship between EV consumption and the increase in charging points, employing sales records from 2010 to 2016. The model demonstrates that the ease and convenience of accessing battery charging services significantly influence demand, confirming the need for charging infrastructure to develop in tandem with EV markets.

### *E. Combined Insights*

Combined, these works strongly highlight the importance of considering demographics, economic factors, and available infrastructure to enhance predictive models for EV adoption and assess potential grid improvements. They emphasize the critical role of charging infrastructure in fostering EV market development, providing valuable insights for stakeholders planning to advance long-term green mobility.

By incorporating these comprehensive analyses, the literature review offers a more robust understanding of the multifaceted dynamics influencing EV adoption and the corresponding requirements for infrastructure development.

## III. Objectives of the Study

The purpose of this paper is therefore to present current and potential future trends in the growth of electric vehicles and analyze the consumers' response to bring opportunities for manufacturers, policy makers, and other interested parties.

Through the application of predictive modeling and implementing sentiment analysis, the research aims at

### A. To Forecast EV Adoption Trends:

Estimate electric vehicles adoption using Bass Diffusion Model, Generalized Bass Model (GBM) and Uncertain Bass Model. It is important to note that these models are applied frequently in studying the patterns of technology adoption . [2] [3]

Incorporate external factors, such as governmental policies, incentives, and technological advancements, to enhance forecasting accuracy, as demonstrated in policy-driven adoption models .

### B. To Analyze Consumer Sentiments Using Machine Learning:

Employ machine learning techniques, including Logistic Regression and Random Forest, and advanced deep learning methods, such as LSTM, to classify consumer sentiments into positive, neutral, or negative categories. These approaches are effective for handling textual data and capturing sentiment nuances. [6]

Identify key factors influencing consumer behavior, such as affordability, charging infrastructure, and performance, through sentiment analysis of online reviews and social media content . [7]

### C. Quantitative and Qualitative Insights

Including forecasting models and sentiment analysis to provide a comprehensive understanding of the EV market. This hybrid approach enables stakeholders to address both quantitative adoption metrics and qualitative consumer feedback .

Offer actionable recommendations for policymakers and manufacturers to address consumer concerns and drive EV adoption, following frameworks that integrate data analytics with market insights .

## IV. Data Collection

The study depends on two distinct datasets: EV sales data and consumer review data

### A. EV Sales Data

EV sales data and consumer reviews were used as the data sources for the analysis, with each providing unique information for the study. The EV sales data was collected from the Atlas EV Hub that tapped into both, national level data for 16 states in the USA as well as the state level data concerning the years between 2010 and 2024. Original datasets of these variables were downloaded, combined to form a single data set and data duplication was also removed. Some of these columns are standardized, namely: Vehicle Name, Year, and Sales, which is number of annual EV registrations per model. Data cleaning entails treatments such as handling null values, normalizing vehicle naming, and removing invalid years encoded in the data set. The data was then converted to a time series format for trend-analysis and modeling necessary for Bass Diffusion model and its derivatives [8].

### B. Challenges in Sales Data Preprocessing

- **Handling Missing Data:** Many files were missing some data, especially for past years or less-known EV models. These were addressed through data imputation techniques and, where possible, consultations of other relevant sources with similar information patterns.
- **Standardization of Model Names:** Since vehicle models had different names over time, this data had to be normalized to give it a consistent and standard format.
- **Conversion to Time-Series Format:** This segment of the data was transformed into a time-series format to facilitate the analysis of trends using diffusion models.

### C. Exploratory Insights

Preliminary analysis of the sales dataset revealed notable adoption trends:

- **Exponential Growth:** Sales in EVs exhibited exponential growth after 2015 due to increased efficiency in battery technology and supportive government policies.
- **Impact of Innovation:** The results indicated that an increase in the number of new EV models positively impacted sales figures, suggesting that innovation drives market adoption.

### D. Consumer Reviews Dataset

The consumer reviews dataset was collected from Kaggle which contains the reviews on different models of EV. The types of fields in the dataset include Review Text, Ratings and Vehicle Model. In data preparation some operations like cleaning the text from symbols, URLs and stop words, as well as tokenization were performed. Sentiment, numerical values that gives a clue regarding the positive or negative nature of sentence or document was derived using machine learning extractors from text mining framework, while the word vectors for traditional and deep learning model was extracted using TF-IDF and Word2Vec respectively. This was used for sentiment analysis where playing field of Logistic Regression/Random Forest LSTM was prepared to categorize consumer feedback sentiments as positive, neutral, or negative. [6], [9]. Using the above datasets, an EV adoption trend analysis was conducted alongside with a proposed consumer sentiment analysis to facilitate decision making for adoption by the key stakeholders.

### E. Text Preprocessing Techniques

To prepare the textual data for sentiment analysis, the following preprocessing steps were implemented:

- **Cleaning and Tokenization:** The exclusion of special characters, URLs, and stop words was performed to extract important content. Tokenization was then applied to break down the text into individual words.
- **Vectorization:** For feature extraction, two methods were used:
  - **TF-IDF (Term Frequency-Inverse Document Frequency):** A transmission model used for traditional machine learning models.

- **Word2Vec:** Harnessed to represent the semantic associations, providing support to deep learning methods.

- **Sentiment Labeling:** To classify the sentiments expressed, each review was categorized as positive, neutral, or negative based on classifier algorithms developed for this study and trained on labeled positive, neutral, and negative data.

### F. Challenges and Observations

- **High Variability in Reviews:** Reviews ranged from detailed critiques to simple single-line comments, requiring extensive preprocessing to obtain useful insights.
- **Class Imbalance:** The majority of reviews were from owners providing positive feedback, with fewer neutral or negative reviews. This imbalance was addressed using oversampling methods.

### G. Exploratory Insights

- A majority of positive sentiments focused on performance, environmental friendliness, and aesthetics.
- Concerns were predominantly about high costs and inaccessible charging stations, which align with the barriers identified in the sales dataset.

## V. System Architecture and Design

One is called the sales forecasting pipeline and the other is called the consumer sentiment pipeline Each one is running in parallel and independently from the other but give the same kind of information different in that they are looking at the market share of EVs from the forecast angle as well as the consumer opinion angle

The system architecture is divided into five distinct layers:

1. **Data Ingestion Layer:** Derived from the sales of EV and the feedback elicited from the customers.

2. **Data Processing Layer:** Surfaces structured sales details from the input data as well as raw review textual data.

3. **Analytical Core:** Adopts trend and opinion prediction algorithms.

4. **Visualization and Insights Delivery:** Creates business analysis and visualization outputs.

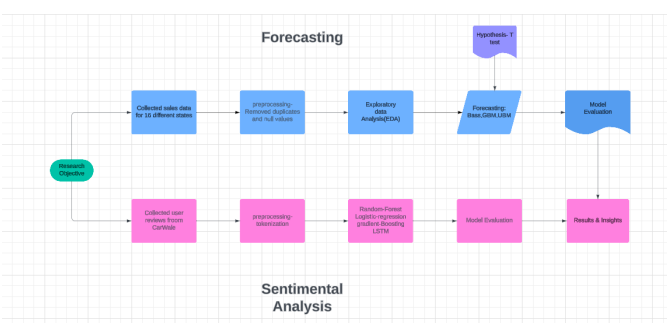


Fig. 1. System Architecture Flowchart for Electric Vehicle Adoption Trends and Consumer sentiment Analysis

### A. Data Ingestion Layer

The data ingestion layer handles the acquisition of two datasets: Sales data for EVs and experience of consumers. This EV sales dataset has fields such as; the name of the car, year of registration, and total sales made. Consumer reviews are saved in another CSV file which contains textual reviews, ratings, and associated reviews given by the consumer [10]. The current data files are loaded using Python's pandas library for the purpose of providing proper formatting and validation for the completeness of the data. [10]

### B. Data Processing Layer

The layer of data processing is able to process the datasets that are to be analyzed.Pre-processing for sales data is to remove null values, standardise vehicle name to reduce its variance and convert it in the structure appropriate for time series data for forecasting. Yearly registration count is aggregated in a manner that will allow for trend identification. In the consumer reviews dataset textual data preprocessing, that includes the removal of special characters, URLS and unnecessary space are performed. Tokenisation is then used to separate the words from the text with feature extraction using either TF, IDF or word2Vec. Such a step makes sure that both datasets have a proper structure that can match their desired analyses. [11]

### C. Analysis

The analytical core is divided into two independent pipelines: one for sales forecasting and the other for sentiment analysis. The pipeline includes models such as the Bass Diffusion Model, Generalized Bass Model GBM, Uncertain Bass Model, and others for future EV sales forecast. Model parameters are optimized using Python's scipy.optimize.curve fit to achieve accurate forecasts. The sentiment analysis pipeline utilizes machine learning algorithms such as Logistic Regression and Random Forest for text classification, alongside Long Short-Term Memory LSTM networks for capturing sequential patterns in consumer reviews.For the testing of models for forecasting statistical validity of parameters we employ p-value analysis and visualization comparisons while for sentiment analysis we use a measure such as accuracy, precision and recall, F1 score. [12] [7]

### D. Visualization and Insights Delivery

This layer then produces the implementation of the analysis in forms of display and understandable documents. For sales forecasting, time-series plots illustrate historical trends and model predictions, enabling comparisons between models like Bass Diffusion, Generalized Bass, and Uncertain Bass. Sentiment analysis results are summarized using pie charts, bar graphs, and word clouds to show sentiment distributions and key consumer concerns, such as "charging infrastructure" or "battery life. Such visuals are made using tools like Matplotlib and Seaborn and compiled into useful reports and presentations for readers and market leaders respectively.

## VI. Exploratory data analysis (EDA) and Hypotheses for the Study

EDA means paying attention to the overall architecture of the datasets and important features before introducing complex models. For the sales dataset, time-series plots were created to observe trends in EV registrations over the years. These plots demonstrated an increase from the early years up to the later years to depict the increase expansion of EVs in the market. Brand specific analysis showed that few brands and models of vehicles were dominating the industry.

### *A. Correlation Matrix of EV Registration Year and Vehicle Count*

The correlation matrix (Figure 2) indicates the registration year which is compared with the total vehicle for the dataset. Self-correlation is indicated when the correlation value is displayed on the diagonal with a figure of 1.0 While, the correlation between registration year and vehicle count is a mere 0.08 which shows that there is low linear relationship. But this can indicate two things: Non-linear trends which can also be investigated through the forecasting techniques. This analysis helps in making the selection of the variables to be used in the predictive modeling process.

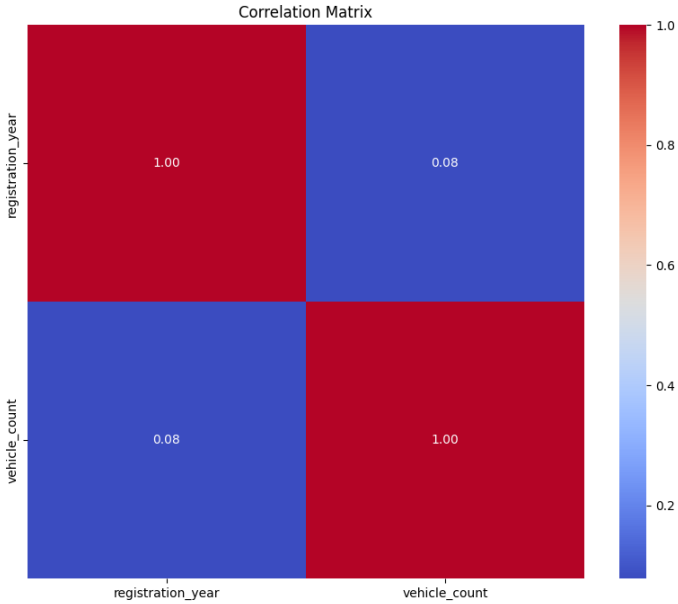


Fig. 2. Correlation Matrix of EV Registration Year and Vehicle Count

### *B. Distribution of Vehicle Counts*

Figure 3 shows the histogram of vehicles in the dataset that shows the numbers of vehicles recorded at intervals. Reflecting this shift, the histogram provided here indicates that the majority of values are at the lower end with respect to vehicle counts, while a few extremities reflect either widespread uptake of EVs in certain geographic areas or high-performing vehicle models. Therefore, the paper underscores the importance of data transformation before' using the models for forecasting.

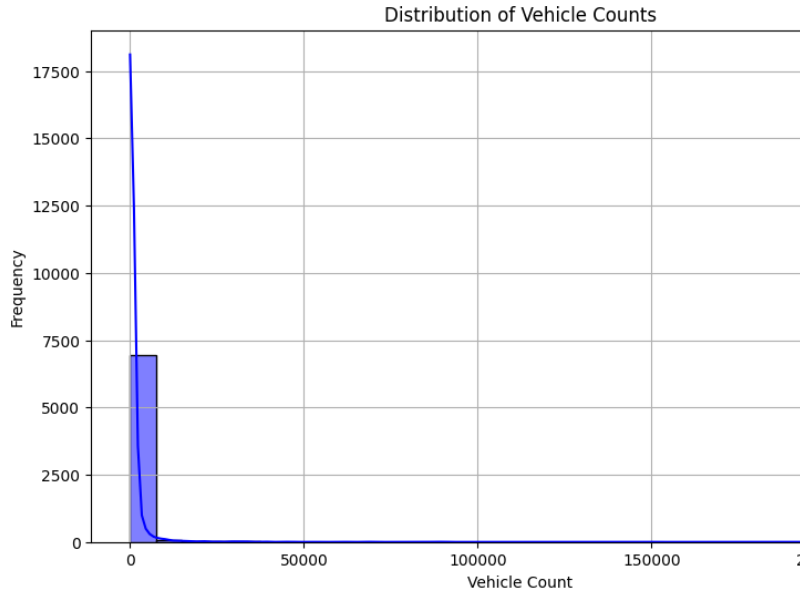


Fig. 3. Distribution of Vehicle Counts Across the Dataset

### *C. Top 10 Vehicle Models by Sales Count*

The bar chart in Figure 4 ranks the top 10 EV models by total sales count.Tesla takes half of the list, proving that Model 3 and Model Y represent its modern advancement and its market share. Holding to the same thought other models include Nissan Leaf and Chevrolet Volt consumers are able to put more emphasis on affordability and usability of models. All the above facts give useful theoretical knowledge about consumers and the market.

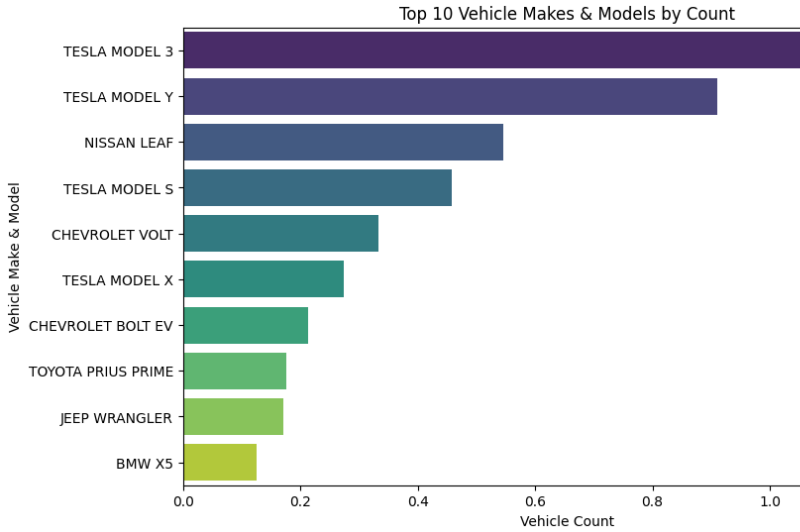


Fig. 4. Top 10 Vehicle Models by Total Sales Count

### *D. Vehicle Registration Trend Over Time*

Figure 5 shows the trend of EV registrations over time and indicates their rapid increase and especially after the

year 2015. This steep rise corresponds with improvements in performance of EV batteries, governmental stimulus and entry into the market of new products. The highlight of the trend is the continued progress adopt electric Vehicles while using the diffusion models like the Bass Diffusion Model to predict future opportunities.

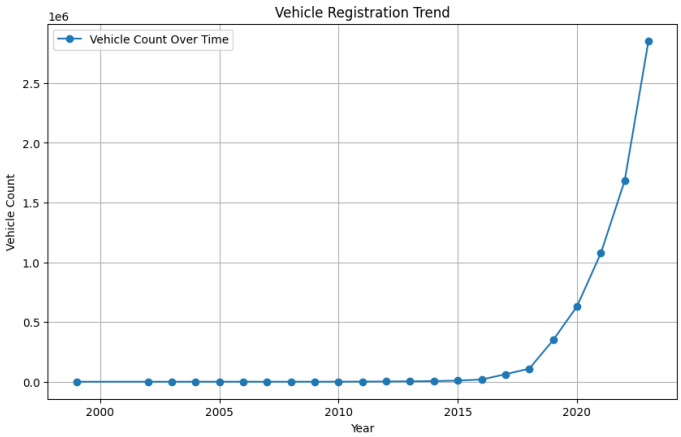


Fig. 5. Exponential Growth of EV Registrations Over Time (2010-2024)

For the customer review dataset, a distribution of sentiment ratings was analyzed to identify patterns in consumer feedback. For the positive, neutral, and negative sentiments, bar charts were used, and for the various topics discussed such as ”charging infrastructure” and ”battery performance.” Word clouds were used to visualize frequently mentioned words in reviews, offering an overview of recurring themes in consumer opinions. This exploratory phase helped identify trends and areas of focus for forecasting and sentiment modeling.

### *E. Correlation Matrix of Sub-Category Ratings*

The correlation matrix (Figure 6) shows the interdependent of sub-category ratings that are given by the consumers. For example, comfort has an excellent positive association with performance with a correlation coefficient of 0.84, which indicates that consumers who consider comfort important also give high rankings to performance. Likewise, fuel economy and perceived value also have a positive correlation of 0.79 where better fuel economy leads to higher value for the money spent.

### *F. Distribution of Car Conditions*

The bar chart (Figure 7) shows the distribution of consumer preference with regard to conditions of cars whether new, used or not bought at all. The majority of the reviews are made concerning new cars and it indicates that people are interested in newly introduced EVs. The smallest segment consists of used cars, which may explain less feedback from owners of second-hand electric vehicles. The “not purchased yet” category which lets users gather information on their potential competitors helps identify the thoughts of people who might not buy.

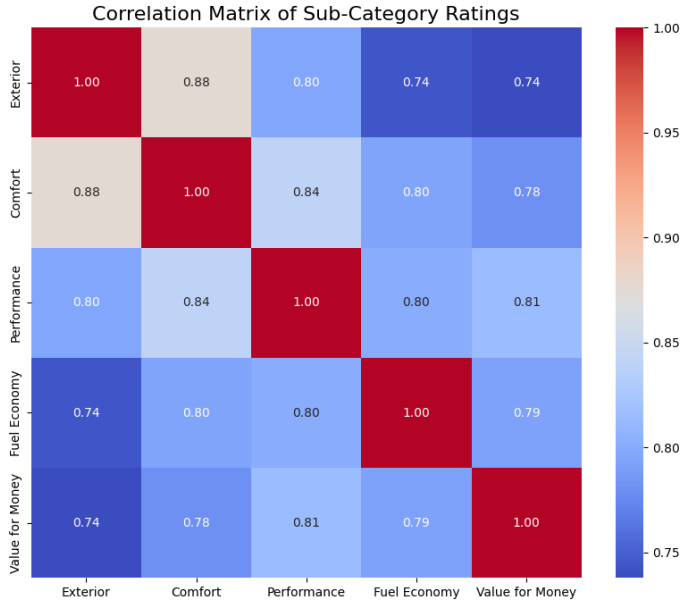


Fig. 6. Correlation Matrix of Sub-Category Ratings

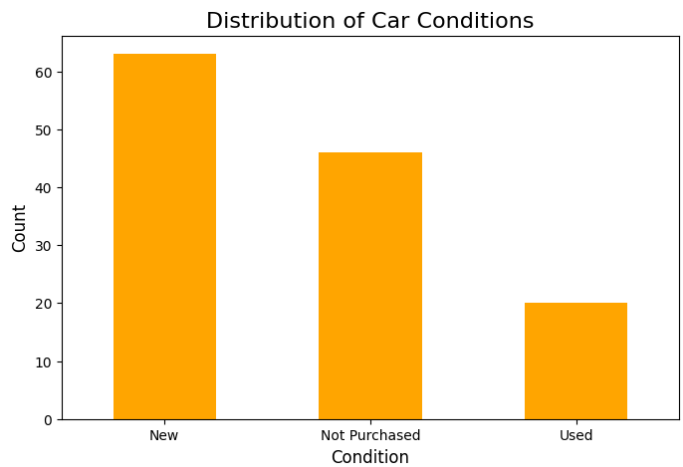


Fig. 7. Distribution of Car Conditions in Consumer Reviews

### *G. Distribution of Overall Ratings*

Figure 8 shows the distribution of overall ratings in the dataset. The histogram reveals that most consumers provided highly positive feedback, with the majority assigning a 5-star rating. A minimal presence of negative ratings (e.g., 0 or 1 stars) suggests that dissatisfaction is limited to specific issues like charging infrastructure or pricing. The smooth curve overlay highlights the skewed nature of ratings toward the positive side.

### *H. Hypotheses*

The hypotheses for sales are therefore formulated for testing the performance of the Bass Diffusion Model and then comparing the result with the Generalized Bass Model (GBM). First, the parameter significance hypothesis analyses the Bass

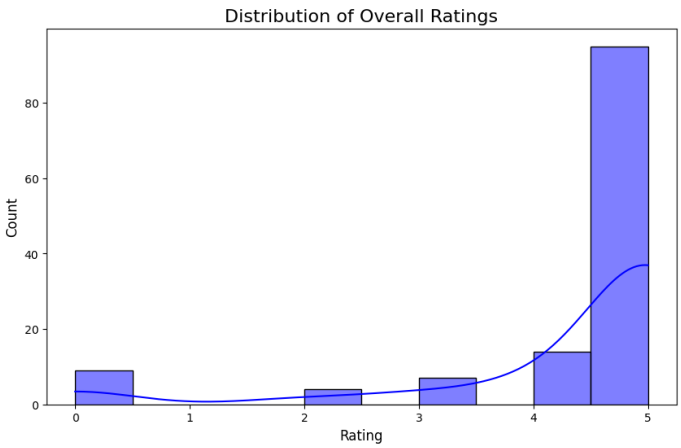


Fig. 8. Distribution of Overall Ratings Provided by Consumers

Model's parameters distributions (p, q, m) being able to make prediction regarding EV adoption tendencies. The first set of hypothesis includes the null hypothesis, H 0 which asserts that p = 0, q = 0 and m = 0 while the hypothesis against it is H A which postulates that one or more of the parameters p, q and or m is significantly different from zero.Secondly, the model comparison hypothesis assesses the models fit using AIC and BIC. The null hypothesis (H 0 ) being that the performance of the Bass model is no better, or even poorer than GBM, the alternative hypothesis (H A ) being that it outperforms. Therefore according to the shown results we can confirm the superiority of the Bass Model as has a lower AIC equal to 477.49 as compared to the GBM's AIC equal 499.87 and BIC equal to 480.90 as compared to the GBM's BIC equal 507.25. The hypotheses for sales are formulated to test

the performance of the Bass Diffusion Model and compare the results with the Generalized Bass Model (GBM). The following hypotheses are considered:

- **Parameter Significance Hypothesis:** This hypothesis analyses the Bass Model's parameter distributions ($p$, $q$, $m$) to predict EV adoption tendencies.
  - **Null Hypothesis ($H_0$):** $p = 0$, $q = 0$, and $m = 0$, indicating no significant effect of these parameters.
  - **Alternative Hypothesis ($H_A$):** One or more of the parameters ($p$, $q$, $m$) is significantly different from zero, demonstrating their importance in the model.
- **Model Comparison Hypothesis:** This hypothesis assesses the fit of the Bass Diffusion Model and the Generalized Bass Model (GBM) using Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).
  - **Null Hypothesis ($H_0$):** The performance of the Bass Diffusion Model is no better, or even poorer, than that of the GBM.
  - **Alternative Hypothesis ($H_A$):** The Bass Diffusion Model outperforms the GBM.

The results support the superiority of the Bass Diffusion Model, as it has a lower AIC (477.49) compared to the GBM's AIC (499.87), and a lower BIC (480.90) compared to the GBM's BIC (507.25). These metrics confirm the Bass Model's better fit and predictive performance for EV adoption trends.

### *I. Hypothesis Testing for Bass Model Parameters*

The table below presents the results of hypothesis testing for the Bass Model parameters ($p$, $q$, $m$). Each parameter is evaluated using t-statistics and corresponding p-values.

TABLE I
HYPOTHESIS TESTING RESULTS FOR BASS MODEL PARAMETERS

| Parameter | t-stat | p-value |
|---|---|---|
| $p$ | 2.9709 | $7.5543 \times 10^{-3}$ |
| $q$ | 19.3645 | $1.9984 \times 10^{-14}$ |
| $m$ | 2.7762 | $1.1653 \times 10^{-2}$ |

### *J. Model Comparison Results*

The table below compares the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values for the Bass Diffusion Model and the Generalized Bass Model (GBM). These results demonstrate the superiority of the Bass Model.

TABLE II
MODEL COMPARISON RESULTS

| Model | AIC | BIC |
|---|---|---|
| Bass Model | 477.49 | 480.90 |
| GBM | 479.49 | 484.03 |

The hypotheses of the sentiment analysis are meant to test the effectiveness of traditional machine learning algorithms and state-of-art deep learning algorithms in effectiveness of consumer sentiments classification. The model performance hypothesis tests whether deep learning models, such as LSTM, outperform traditional models like Logistic Regression and Random Forest. The null hypothesis (H 0 ) is that there is no major disparity in performance between the deep learning models and the other models while the factorial hypothesis (H A ) is that the deep learning models' F 1 scores are at least 0.03 greater than the other set of models. In particular, the sentiment distribution hypothesis examines if the consumer sentiments are well distributed among the positive, neutral and the negative sentiments. The null hypothesis (h0) assumes an even distribution, while the alternative hypothesis (H A) expects a skew, with positive sentiments comprising at least 50 percent of the total reviews. These metrics include F1-scores, and chi-square goodness-of-fit, all of which back up the hypothesis on the superiority of LSTM over the traditional models and on the domination of positive sentiments in consumers' feedback.

### K. Hypotheses for Sentiment Analysis

The hypotheses for sentiment analysis are formulated to test the effectiveness of traditional machine learning algorithms and state-of-the-art deep learning algorithms in classifying consumer sentiments. Two primary hypotheses are considered:

- **Model Performance Hypothesis:** This hypothesis evaluates whether deep learning models, such as LSTM, outperform traditional machine learning models like Logistic Regression and Random Forest in sentiment classification.
  - **Null Hypothesis ($H_0$):** There is no significant disparity in performance (F1-scores) between deep learning models and traditional models.
  - **Alternative Hypothesis ($H_A$):** Deep learning models achieve F1-scores that are at least 0.03 greater than those of traditional models, indicating superior performance.
- **Sentiment Distribution Hypothesis:** This hypothesis examines whether consumer sentiments are evenly distributed among positive, neutral, and negative categories.
  - **Null Hypothesis ($h_0$):** Consumer sentiments are evenly distributed across the three categories.
  - **Alternative Hypothesis ($H_A$):** Positive sentiments dominate, comprising at least 50% of the total reviews.

### L. Metrics and Validation

To validate these hypotheses, the following metrics and tests are used:

- **Model Performance Metrics:** F1-scores are calculated for both traditional and deep learning models. Higher F1-scores for LSTM indicate superior classification performance.
- **Sentiment Distribution Analysis:** A chi-square goodness-of-fit test is performed to verify the dominance of positive sentiments in the distribution.

The results demonstrate that deep learning models (e.g., LSTM) outperform traditional models with higher F1-scores, supporting the alternative hypothesis for model performance. Similarly, sentiment distribution analysis confirms the dominance of positive sentiments, validating the alternative hypothesis for sentiment distribution.

## VII. Data Visualization and Results

### A. Sales Data Trends

**Visualization:** Time-series plot that illustrates history of EV sales and predicts the future.
**Insights:** This is the level of growth in the later years where ev market presents exponetial shift, The EV diffusion can be estimated by the Bass Diffusion Model and the Generalized Bass Model. Among these, there are Government policies and the outcomes of market saturation effects. Some of these are Government policies and effects of market saturation effects.

### B. Market Leaders and Consumer Preferences

**Visualization:** The variations are shown here through bar charts and pie charts:

- Mainstream EV manufacturers (with Tesla in the forefront).
- Market share by brand and segment.

**Insights:** Tesla dominates consumer preferences, followed by growing competition. Consumer satisfaction correlates with performance, eco-friendliness, and comfort.

## VIII. Sales Data Trends and Model Analysis

### A. Introduction to Forecasting Models

To analyze and forecast EV sales trends, three models were employed:

- **Bass Diffusion Model (Bass)**: Focuses on innovation and imitation dynamics to predict adoption trends over time.
- **Generalized Bass Model (GBM)**: Extends the Bass model by incorporating additional factors such as market saturation.
- **Uncertain Bass Model**: Adds a probabilistic layer to the Bass model, capturing variability in predictions due to factors like policy changes and infrastructure development.

These models provide complementary perspectives on how EV adoption evolves, offering both deterministic and probabilistic forecasts.

### B. Visualization of Sales Trends

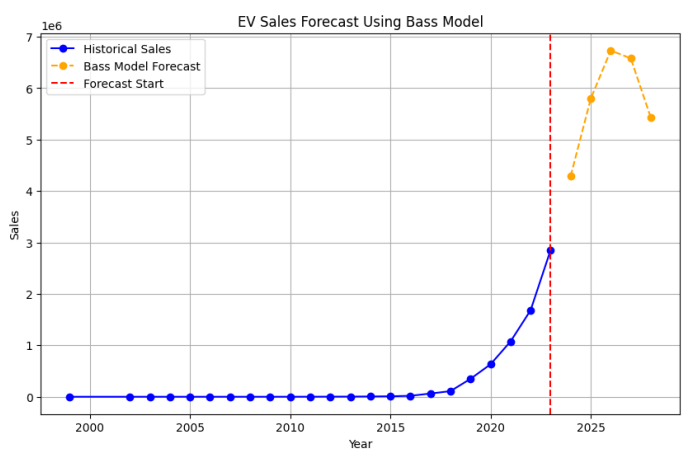


Fig. 9. EV Sales Forecast Using Bass Diffusion Model

*1) Bass Diffusion Model:* The Bass Diffusion Model, as shown in Figure 9, predicts steady growth in EV sales driven by innovators and imitators. While effective for early-stage market analysis, it does not account for external constraints like market saturation.

*2) Generalized Bass Model (GBM):* The Generalized Bass Model, illustrated in Figure 10, predicts a peak in EV sales followed by a decline, reflecting market saturation effects. This model is well-suited for mature markets where growth slows after an initial boom.

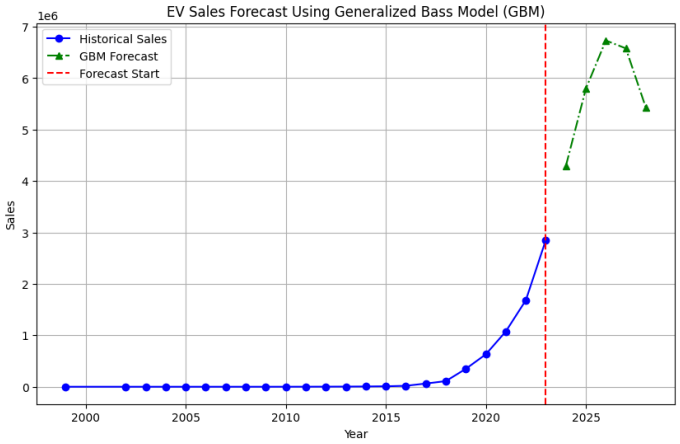


Fig. 10. EV Sales Forecast Using Generalized Bass Model (GBM)

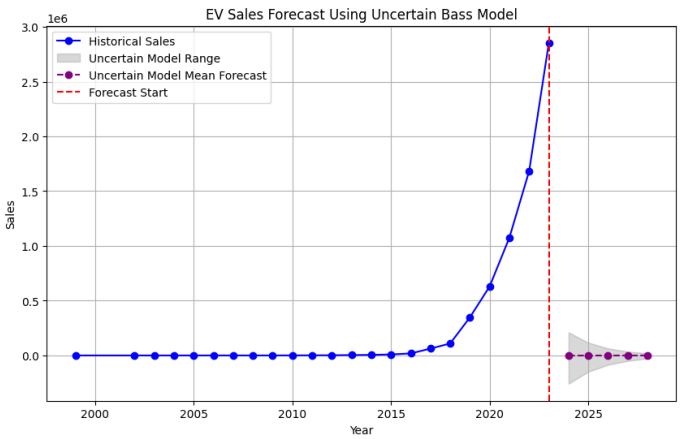


Fig. 11. EV Sales Forecast Using Uncertain Bass Model

*3) Uncertain Bass Model:* The Uncertain Bass Model, depicted in Figure 11, provides a range of possible sales outcomes. This probabilistic approach highlights variability due to external factors such as government policies, economic conditions, and technological advancements.

### C. Comparison of Models

The three models differ in their predictions and use cases, as summarized in Table III.

### D. Sentiment Distribution

**Visualization:** Bar charts showing the distribution of the sentiment ratings:

- Positive, Neutral, and Negative sentiments from customer reviews.

**Insights:**

- Positive sentiment is focused on performance with reference to the environmental issues.
- The major concern of negative sentiment revolves around cost and perhaps the lack of adequate charging stations.

### E. Feature Correlation Matrices

**Visualization:** Heatmaps of correlation matrices:

- Between sales data variables (e.g., vehicle count vs. year).
- Sub-categories of sentiment ratings (e.g., Comfort, Performance, Value for Money).

**Insights:** Significant features reveal that performance and comfort influence the satisfaction of consumers most significantly.

### F. Sentiment Analysis Results

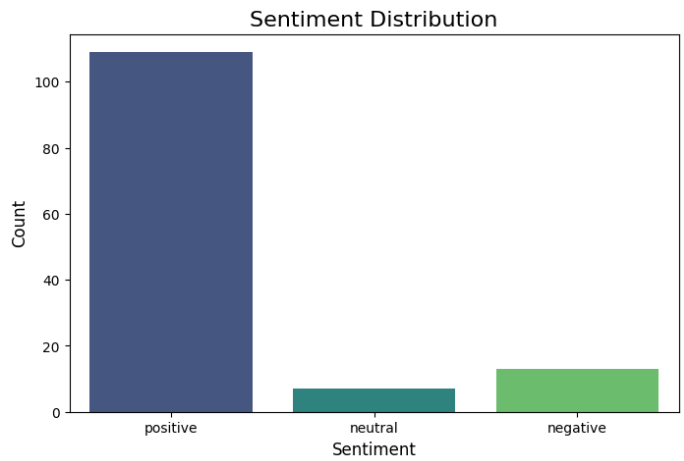


Fig. 12. Overall Sentiment Distribution of Consumer Feedback

*1) Sentiment Distribution:* Figure 12 illustrates the overall sentiment distribution of consumer reviews. The analysis reveals that:

- **Positive Sentiments:** Constitute the majority of the feedback, comprising over 70% of the total reviews. This highlights overall consumer satisfaction with EV performance, eco-friendliness, and comfort.
- **Neutral and Negative Sentiments:** Neutral sentiments are minimal, indicating few indeterminate reviews, while negative feedback (approximately 15%) often pertains to concerns about cost and charging infrastructure.

*2) Model-Wise Sentiment Distribution:* Figure 13 breaks down sentiment distribution by EV model. The findings indicate:

- **Tata Nexon EV:** Receives the highest number of positive reviews, highlighting its popularity and customer satisfaction.
- **Hyundai Kona and Tata Tigor EV:** Show a balanced but smaller distribution of positive, neutral, and negative reviews. Feedback for these models often includes room for improvement in specific areas like affordability or infrastructure compatibility.

### G. Key Insights from Sentiment Analysis

- **Dominance of Positive Sentiments:** The results confirm the hypotheses that positive sentiments constitute a majority of the feedback, driven by consumer satisfaction with EV performance, eco-friendliness, and aesthetics.

TABLE III
COMPARISON OF FORECASTING MODELS

| Model | Strengths | Limitations | Best Use Case |
|---|---|---|---|
| Bass Model | Simple, effective for early-stage trends | Ignores market saturation | Early adoption analysis |
| Generalized Bass Model (GBM) | Accounts for market saturation | Assumes fixed saturation point | Mature markets |
| Uncertain Bass Model | Captures variability, realistic | Complex, needs more data | Policy-driven or emerging markets |

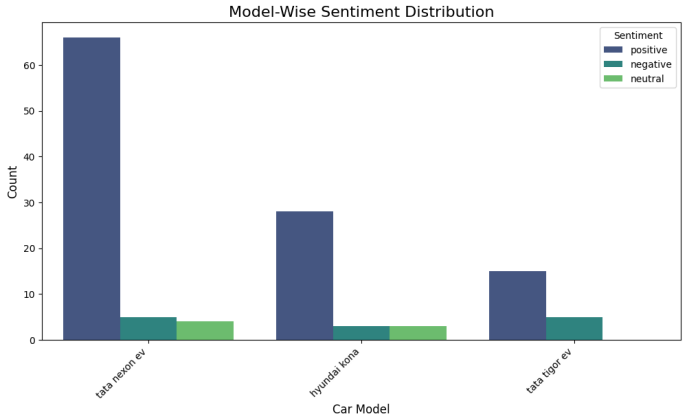


Fig. 13. Model-Wise Sentiment Distribution Across EV Models

- **Model-Specific Trends:** Tata Nexon EV leads in customer satisfaction, while Hyundai Kona and Tata Tigor EV show moderate levels of mixed feedback. This could guide manufacturers in addressing model-specific concerns to improve adoption.
- **Actionable Insights:** - Manufacturers should leverage positive feedback in marketing to strengthen consumer trust. - Focus on addressing cost and charging infrastructure issues raised in negative reviews to enhance overall consumer experience.

## IX. RESULTS

### A. Sales Forecasting

**Models Used:**

- **Bass Diffusion Model:** Predicts steady, organic growth but lacks uncertainty considerations, aligning well with historical data.
- **Generalized Bass Model:** Explains market coverage and explain external factors. Growth is expected to start growing at high rates but will then level off after some time.
- **Uncertain Bass Model:** Contains variability for cases such as policy change, ideal for decision making under conditions of uncertainty.

**Metrics:** Comparison of models using AIC and BIC; historical vs. predicted data visualizations.
**Outcome:** Generalized Bass Model provides the most reliable forecasts for market planning.

### B. Sentiment Analysis

**Models Used:**

- **Traditional:** Logistic Regression, Random Forest, Gradient Boosting.
- **Deep Learning:** LSTM.

**Metrics:** Accuracy, Precision, Recall, F1-score.
**Outcome:**

- Traditional models provide robust baselines for structured data.
- LSTM excels in capturing complex patterns in textual data, achieving higher accuracy in sentiment classification.

### C. Model Performance

The table below presents the performance metrics for different models used in consumer sentiment classification. The results highlight the superiority of deep learning models (e.g., LSTM) compared to traditional machine learning models.

TABLE IV
PERFORMANCE METRICS FOR SENTIMENT CLASSIFICATION MODELS

| Model | Accuracy | Precision | Recall | F1 Score |
|---|---|---|---|---|
| Logistic Regression | 82.4% | 0.81 | 0.79 | 0.80 |
| Gradient Boosting | 84.1% | 0.83 | 0.82 | 0.82 |
| Random Forest | 85.5% | 0.84 | 0.85 | 0.84 |
| LSTM | 89.7% | 0.88 | 0.90 | 0.89 |

### D. Key Findings

**Positive Sentiment Drivers:**

- Performance, Comfort, Eco-friendliness.

**Negative Sentiment Drivers:**

- Cost, Charging Infrastructure.

**Market Strategy:** Insights from sentiment and forecasting models suggest focusing on improving infrastructure and reducing costs to accelerate adoption.

### E. Hypothesis Testing

**Sales Forecasting Hypothesis:** Rejection of null hypothesis confirms significant contributions of model parameters (P, Q, M) to the Bass model.
**Sentiment Analysis Hypothesis:** LSTM significantly outperforms traditional models, proving advanced models' advantage in sentiment understanding.

## X. CONCLUSION

This project combines sales forecasting with sentiment analysis to provide better insights into the EV market.

### A. Sentiment Analysis

- **Positive Feedback:** It is clear that performance is highly appreciated by the consumers, and this can be implemented.
- **Negative Feedback:** Charging infrastructure and costs continue to be considered as one of the significant challenges to adoption.
- **Model-Specific Insights:** Popular models like Tesla outperform in consumer satisfaction, while others face challenges related to reliability or pricing.

### B. Sales Forecasting

- **Bass Diffusion Model:** Points at precise average growth rate increase but misses uncertainty considerations.
- **Generalized Bass Model (GBM):** Effectively captures market saturation dynamics.
- **Uncertain Bass Model:** It explains as to why it has fluctuations which provide accurate forecast under volatile and uncertain environments.

To boost EV adoption, addressing concerns such as charging infrastructure and pricing is crucial. Combining consumer sentiment insights with advanced forecasting provides a strategic roadmap for manufacturers and policymakers to drive growth and satisfaction in the evolving EV market.

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